AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2019

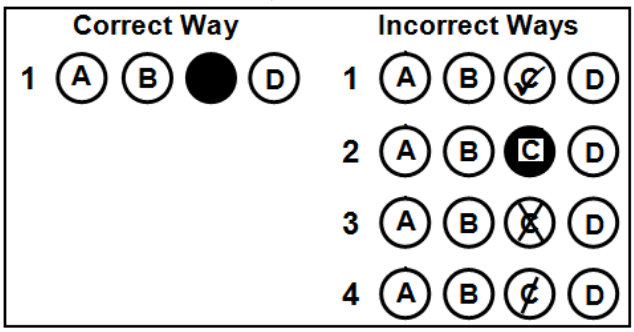
Biology Paper I

Time: 40 minutes Marks: 25

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INSTRUCTIONS

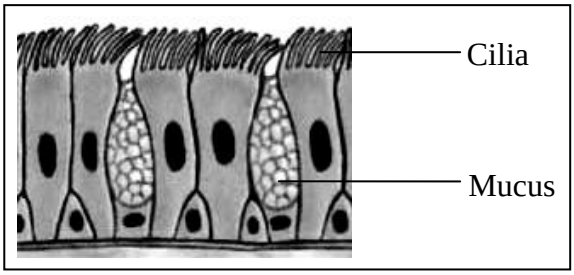
1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question, there are four choices A, B, C, D. Choose ONE. On the answer grid, black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. The given diagram shows a layer of cells found in the trachea of human beings.



The function of these cells is to

A. exchange gases.
B. filter the inhaled air.
C. prevent the entry of food in trachea.
D. prevent the collapse of tracheal walls.

2. The given table shows the percentage of oxygen gas in inspired and expired air.

Oxygen in	
Inspired Air (%)	**Expired Air (%)**
21.0	16.4

This difference in the amount of oxygen gas is due to the

A. storage of oxygen gas in lung tissues.
B. non-reactivity of oxygen gas with the blood.
C. liberation of oxygen gas as a metabolic waste.
D. utilisation of oxygen gas in energy production.

3. In asthma, shortness of breath occurs due to

A. constriction of bronchi.
B. rupturing of alveolar walls.
C. rupturing of bronchial walls.
D. accumulation of pus in alveoli.

4. In the kidneys of human beings, blood pressure plays an important role in the process of urine formation. This is because blood pressure helps to

A. maintain salts in the filtrate.
B. form filtrate in the glomerulus.
C. secrete hydrogen ions in tubules.
D. reabsorb glucose in the Loop of Henle.

5. Lithotripsy is a procedure that breaks large kidney stones into smaller ones by using

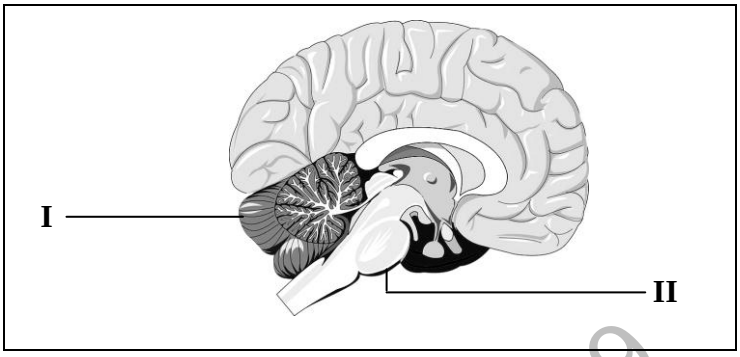
A. X-rays.
B. microwaves.
C. sound waves.
D. ultraviolet rays.

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6. In contrast to nerve cells, the response produced by hormones takes longer because

A. hormones are specific in their action.
B. hormones are transported through blood.
C. hormones are produced in small amounts.
D. the molecules of hormones are small in size.

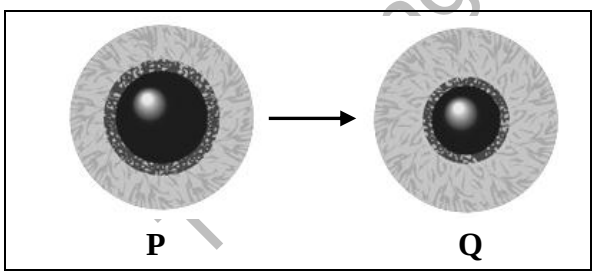
7.



The parts of brain represented by **I** and **II** are

	I	II
A	cerebellum	pons
B	cerebrum	pons
C	cerebellum	medulla oblongata
D	cerebrum	medulla oblongata

8. The given diagram shows change in the pupil of a person from state **P** to **Q**.



The state **Q** shows that the circular muscles of the iris have

A. relaxed as the person entered a poorly lit room.
B. relaxed as the person entered a brightly lit room.
C. contracted as the person entered a poorly lit room.
D. contracted as the person entered a brightly lit room.

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9. The bone that belongs to the appendicular skeleton of human beings is

   A. femur.
   B. stapes.
   C. sternum.
   D. hyoid bone.

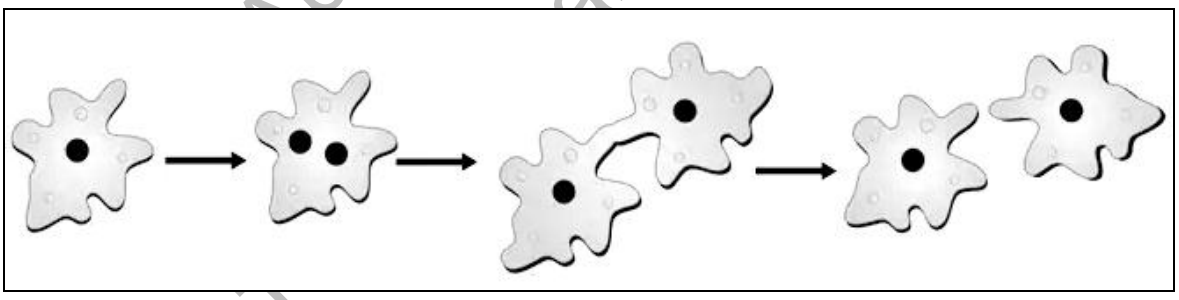
10. Which of the following is the condition of biceps during flexion and extension in the arm of human beings?

	The Condition of Biceps	
	during Flexion	**during Extension**
A	Relax	Relax
B	Contract	Relax
C	Relax	Contract
D	Contract	Contract

11. Osteoporosis characteristically occurs in elderly women due to the deficiency of

   A. thyroxin.
   B. oxytocin.
   C. oestrogen.
   D. calcitonin.

12. The given diagram represents reproduction in an organism.



In the given case, the offspring lack genetic variation because they

   A. are limited in number.
   B. are produced asexually.
   C. are unicellular organisms.
   D. contain a single chromosome.

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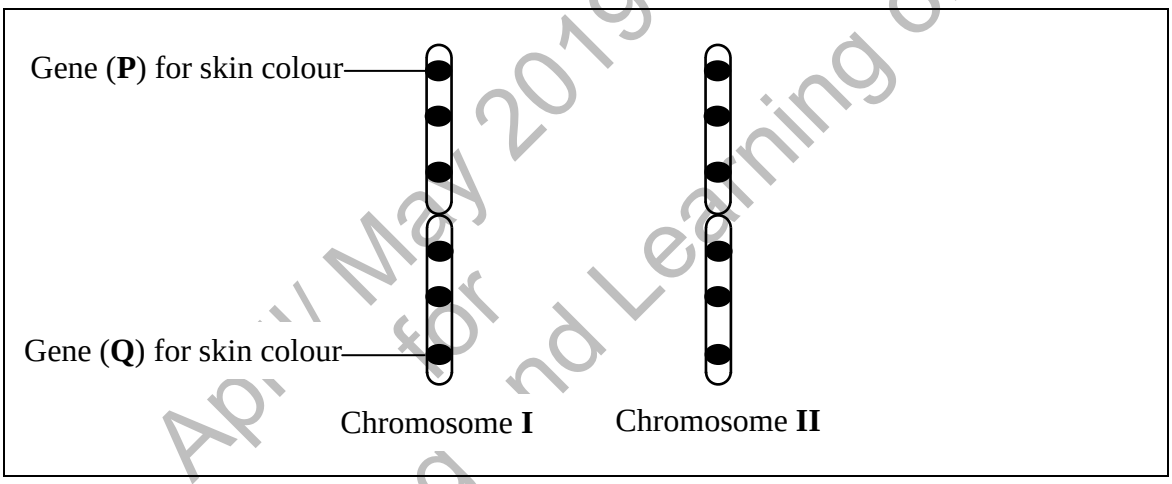
13. During the process of spermatogenesis in male rabbit, a spermatogonium undergoes mitosis to produce

    A. sperms.
    B. spermatids.
    C. primary spermatocytes.
    D. secondary spermatocytes.

14. Human immune deficiency virus (HIV) causes acquired immune deficiency syndrome (AIDS) in human beings by

    A. increasing the pH of blood.
    B. decreasing white blood cells.
    C. increasing the division of cells.
    D. decreasing the oxygen carrying capacity of blood.

15. The given diagram shows genes on a pair of homologous chromosomes.

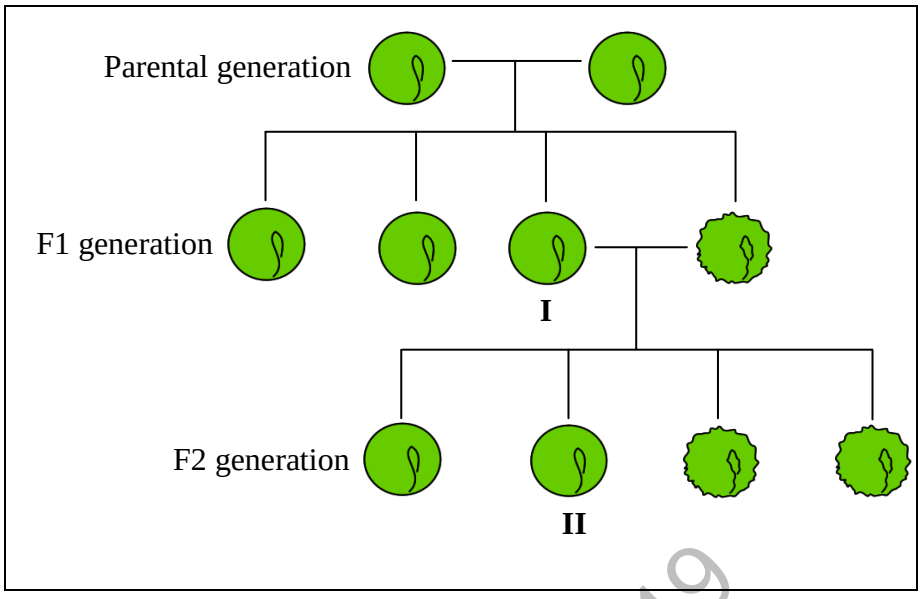


Allele for the gene **P** on chromosome **II** of the given homologous pair could be

    A. P or p.
    B. P or q.
    C. P only.
    D. Q only.

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16. The given diagram shows a genetic cross between two round pea seeds.

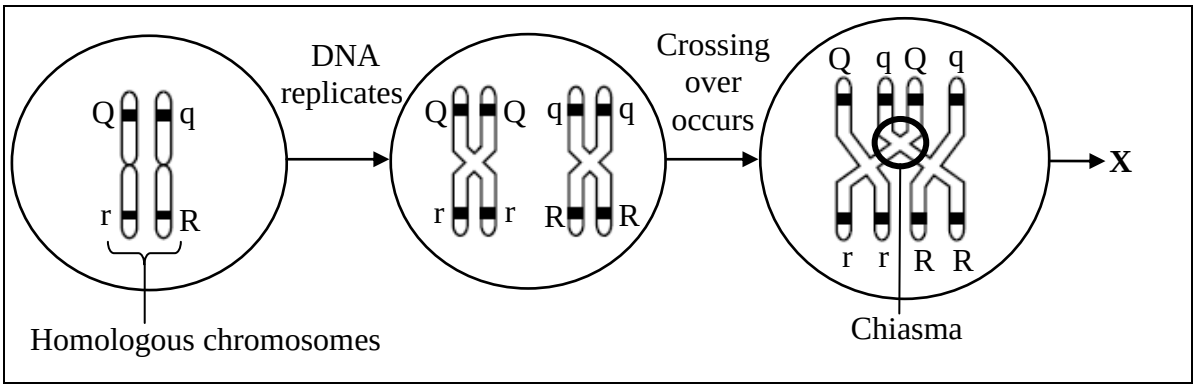


The genotypes of seed **I** and **II** would be

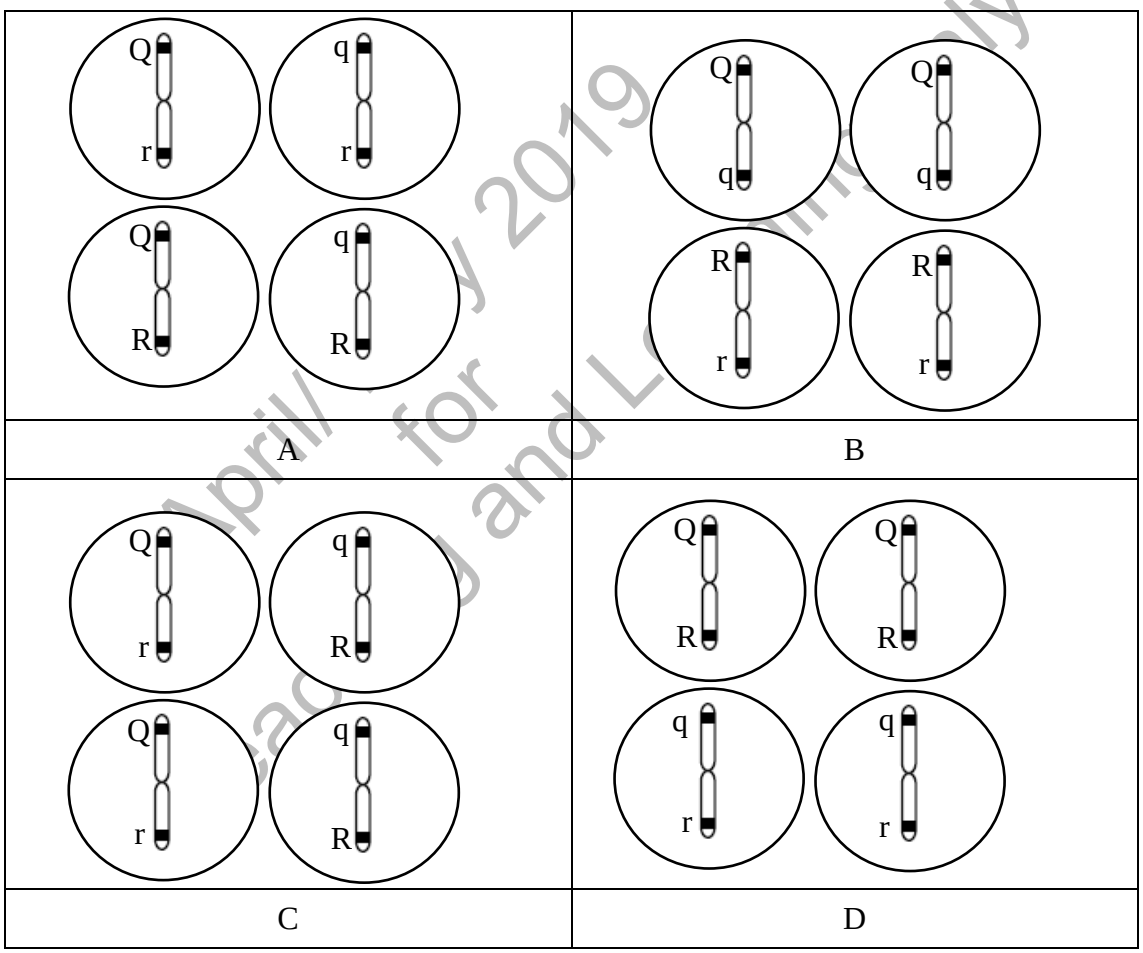
	Genotype of I	**Genotype of II**
A	homozygous	homozygous
B	homozygous	heterozygous
C	heterozygous	homozygous
D	heterozygous	heterozygous

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17. The given diagram illustrates the Mendel's law of independent assortment.



After the completion of meiosis, the types of gametes formed at **X** would be



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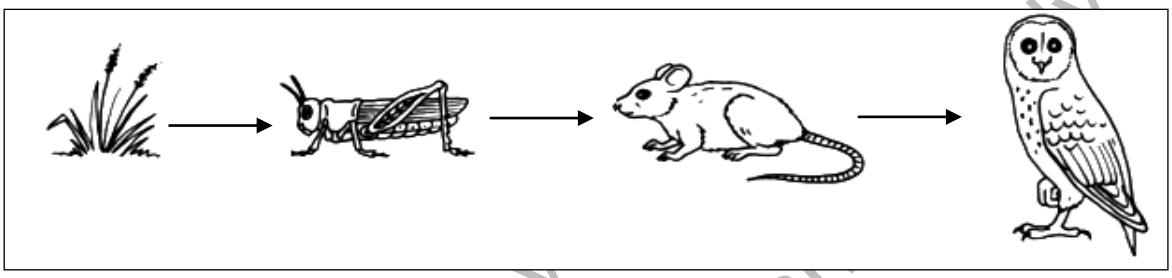
18. Which of the following is an example of artificial selection?

A. Grapes are bred to produce large berry size.
B. Ladybirds are used to control aphids on crops.
C. Bacteria develop strains that are resistant to antibiotics.
D. Human gene is inserted into bacteria to produce insulin.

19. The function of fungi in an ecosystem is to

A. provide oxygen gas to producers.
B. return nutrients to the environment.
C. increase complexity of food chains.
D. decrease competition among consumers.

20. In the given food chain, the trophic level that contains the largest biomass would be at



A. rat.
B. owl.
C. grass.
D. grasshopper.

21. Barnacles are a type of sea animals that attach to whales. Barnacles cannot move on their own; therefore, they use the whale to move around and find locations with food.

The type of animal association that best fits in the given situation is

A. predation.
B. parasitism.
C. mutualism.
D. commensalism.

22. Interferons are special proteins produced by human cells that

A. reduce pain.
B. cure thalassemia.
C. dissolve blood clots.
D. limit spread of viral infections.

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23. Which of the following CORRECTLY matches the drug with its source?

	Drug	Source of Drug
A	Streptomycin	Microorganism
B	Morphine	Animal
C	Penicillin	Plant
D	Aspirin	Mineral

24. The drugs that are classified as medicinal drugs are

A. antibiotics and heroin.
B. caffeine and antibiotics.
C. anaesthetics and caffeine.
D. anaesthetics and analgesics.

25. Bacteriostatic antibiotics work by

A. killing bacteria.
B. inhibiting growth of bacteria.
C. developing resistance in bacteria.
D. producing immunity against bacteria.

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END OF PAPER

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